

IDEA TITLE

# VARSHASETU: Hyperlocal Monsoon Onset, Break & Agricultural Advisory System

Bridging Subseasonal Climate Intelligence with Every Smallholder Farmer at Block & Village Cluster Scale

PROBLEM STATEMENT ID

SIH26086

PROBLEM STATEMENT TITLE

Hyperlocal Monsoon Onset & Break Prediction System (Block/  
Village Scale)

MINISTRY / ORGANIZATION

Ministry of Earth Sciences (MOES) / NCMRWF



## Proposed Solution: VarshaSetu

A Subseasonal-to-Seasonal (S2S) AI/ML platform transforming coarse climate data into village-level actionable advisories.

### Detailed Explanation

- **Hyperlocal S2S Forecasting:** Predicts 7–28 day multi-horizon rainfall at 5–10 km Block & Village cluster scales.
- **Monsoon Onset Assessment:** Evaluates adapted IMD thermodynamic criteria (Rainfall  $\geq 2.5$ mm, 850hPa wind  $\geq 7.7$ m/s, OLR  $\leq 200$ W/m<sup>2</sup>).
- **Intra-Seasonal Spell Outlooks:** Detects active wet spells and severe break dry spells (5–7+ days) 10–14 days in advance.
- **ICAR-CRIDA Agro-Engine:** Generates dynamic stage-specific guidance for 6 major Kharif crops (Paddy, Cotton, Soybean, Groundnut, Maize, Redgram).

### Addressing the Problem

- **Eliminates Scale Mismatch:** Replaces coarse 50–100 km district forecasts that mask micro-climate rainfall variations.
- **Prevents Re-sowing Disasters:** Warns farmers against premature sowing before 2-week dry break spells cause seed desiccation.
- **Protects Fertilizer Inputs:** Prevents nitrogenous top-dressing washout by alerting against heavy rain events ( $>64.5$ mm/day).
- **Dual-Workflow Design:** Delivers simple micro-actions to farmers and mandal-level aggregate tactical directives to Extension Officers.

### Innovation & Uniqueness

- **Physics-Informed Hybrid ML:** Couples global teleconnections (NOAA ENSO ONI, IOD, MJO) with regional agro-meteorological telemetry.
- **Contingent Crop-Choice Engine:** Suggests short-duration, drought-hardy cultivars (e.g., Telangana Sona RNR 15048, Kadiri-6) if onset is delayed  $>14$  days.
- **Inclusively Multilingual:** 7 Indian languages (EN, TE, HI, TA, KN, UR, ML) featuring native Urdu RTL support.
- **Web Speech Audio TTS:** One-tap vernacular text-to-speech narration overcoming rural illiteracy barriers.

## Technical Stack & Implementation Architecture

Modular micro-services architecture combining physics-informed machine learning with low-bandwidth GIS and rural communications.

### BACKEND

#### Python 3.14 & Flask

WSGI REST APIs, Blueprint routing

### ML ENGINE

#### Scikit-Learn & Joblib

Random Forest, 18 S2S features

### GIS & WEB UI

#### Leaflet.js & Chart.js

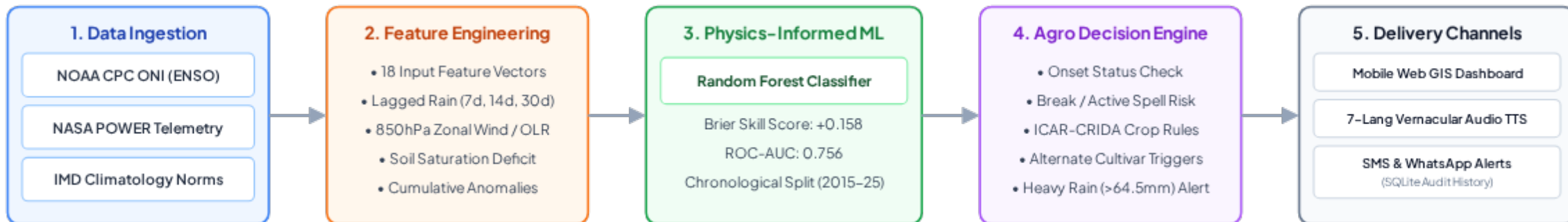
Village polygons, NLM isochrones

### MESSAGING & AUDIT

#### SQLite & Web Speech

7-Lang i18n, SMS/WhatsApp sandbox

### SYSTEM IMPLEMENTATION PIPELINE & DATA FLOW



# Feasibility, Model Metrics & Risk Mitigation

Rigorous chronological model validation backed by transparent risk management strategies.

BRIER SKILL SCORE

**+0.158**

+15.8% Skill Over Climatology

ROC-AUC SCORE

**0.756**

Strong Spell Discrimination

AUTOMATED TESTS

**60 / 60**

Pytest Backend Suite Passed

MOBILE VIEWPORTS

**42 / 42**

100% Mobile Audit Passed

| Risk & Challenge Analysis Matrix       |                                                                                         | Zero Temporal Leakage Policy (Chronological Train: 2015-22   Test: 2023-25)                       |
|----------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Feasibility Dimension                  | Potential Challenge / Risk                                                              | Mitigation Strategy Implemented                                                                   |
| Live NCMRWF Upper-Air Data Feed Access | NCMRWF ERPS socket feeds require MoES high-security institutional credentials.          | Calibrated climatological benchmark distributions used as fallback & labeled explicitly in UI.    |
| Atmospheric Chaos Beyond 14 Days       | Deterministic weather prediction degrades on 21–28 day subseasonal horizons.            | Employs S2S probabilistic weekly anomaly distributions with expanding uncertainty bounds.         |
| Rural Connectivity & Literacy Gaps     | Intermittent 2G networks and inability of marginal farmers to read technical forecasts. | Local SQLite caching, 7-lang Web Speech TTS audio narration & sandboxed SMS/ WhatsApp dispatches. |
| Data Provenance & Misrepresentation    | Prototype systems hiding whether data is real, hardcoded, or simulated.                 | 5-Tier Provenance Architecture explicitly declaring real APIs vs benchmark fallbacks in UI.       |

# Socio-Economic Impact & Multi-Stakeholder Benefits

Transforming agricultural resilience across India's rainfed farming belts.

## Impact on Smallholder & Marginal Farmers

- **Zero-Jargon Actionable Guidance:** Simple plain-language advice (Sow, Delay, Irrigate, Hold Fertilizer) delivered directly in native dialects.
- **Audio Accessibility:** Text-to-speech audio narration enables illiterate farmers to listen to advisories on budget smartphones.
- **Re-sowing Cost Protection:** Timely break-spell warnings prevent seed mortality, protecting smallholders from crippling re-sowing debt cycles.
- **Resilient Cultivar Adoption:** Automatic guidance to pivot to short-duration alternative crops (e.g., Telangana Sona RNR 15048).

## Impact on Extension Officers (AEOs/KVK Scientists)

- **Mandal Spatial Oversight:** Interactive Leaflet map displaying block vulnerability clusters (Low, Moderate, High, Severe).
- **Contingency Seed Planning:** Triggers early mobilization of drought-resilient buffer seed stocks when prolonged dry spells are detected.
- **Input Supply Synchronization:** Directs credit societies (PACS) to withhold subsidized urea ahead of heavy rain events to prevent runoff.
- **Mass Broadcast Dispatch:** Synchronized SMS/WhatsApp broadcasting to mandal-wide subscriber lists with full audit logs.

## Economic Benefits

Saves **₹5,000–₹8,000 per acre** in avoided re-sowing expenses and wasted fertilizer inputs. Optimizes crop yield potential across 55% rainfed sown area.

## Social Benefits

Mitigates climate distress among 600M+ citizens. Democratizes high-end S2S climate science into accessible vernacular audio for rural equity.

## Environmental Benefits

Prevents chemical fertilizer leaching into aquatic ecosystems during extreme rainfall. Conserves agricultural groundwater reserves via smart irrigation timing.

## Research Framework, Citations & Scientific References

Grounded in operational meteorological protocols and agricultural research standards.

### Meteorological & Operational Protocols

- **India Meteorological Department (IMD):** Operational criteria for Southwest Monsoon onset declaration & subseasonal forecasting framework (Rainfall, 850hPa Zonal Wind, OLR).
- **NCMRWF & MoES:** National Centre for Medium Range Weather Forecasting Extended Range Prediction System (NEPS/ERPS) reanalysis protocols.
- **NOAA Climate Prediction Center (CPC):** Oceanic Niño Index (ONI) 3-month running mean SST anomalies in the Niño 3.4 region (5°N–5°S, 120°–170°W).
- **Australian Bureau of Meteorology (BOM):** Indian Ocean Dipole (DMI) & Wheeler-Hendon Real-time Multivariate MJO (RMM1/RMM2) index benchmarks.

### Agronomic Rules & ML Verification Standards

- **ICAR - CRIDA Protocols:** Indian Council of Agricultural Research & Central Research Institute for Dryland Agriculture technical bulletins & contingent crop planning guidelines.
- **S2S Predictability & Brier Skill Metrics:** Chronological out-of-sample cross-validation standards for subseasonal time-series data without future temporal leakage.
- **NASA POWER Agro-Climatology:** Surface agro-meteorological observations and satellite solar energy parameters.
- **W3C Web Speech & Telecom DLT:** BCP-47 localized voice synthesis standards & TRAI DLT template formatting guidelines.

### Prototype Ready for MOES / NCMRWF Pilot Deployment

VarshaSetu is a fully functional working prototype with 60/60 Pytest backend tests passed, verified SQLite notification audit logging, 7-language dynamic DOM translation, and responsive Leaflet GIS mapping.

SIH26086 COMPLIANT